

TECHNICAL REPORT

Secure Decentralized Drone Swarm Management System with ML-Driven Obstacle Avoidance, Acoustic Source Localization, and Blockchain-Based Flight Ledger

Md Shahanur Islam Shagor

Project Architect & Lead Developer

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Abstract

This paper presents a comprehensive engineering framework for a secure, decentralized, autonomous drone swarm management system specifically designed to operate in GPS-denied, acoustically complex, and adversarially contested environments. Contemporary swarm systems suffer from three fundamental vulnerabilities: (i) single-point-of-failure centralized control architecture susceptible to Byzantine faults; (ii) overreliance on GPS and optical sensors that degrade catastrophically in low-visibility conditions such as fog, smoke, or electronic warfare jamming; and (iii) absence of immutable telemetry integrity mechanisms, exposing data streams to spoofing and replay attacks. Our proposed system addresses these deficiencies through six tightly-integrated technical subsystems: a multi-criteria weighted leader election protocol based on the Bully algorithm with battery, motor health, and communication stability scoring; AES-256-GCM encrypted peer-to-peer mesh communication over UDP multicast without infrastructure dependency; an ML-driven dynamic obstacle avoidance engine employing collision-cone probability modeling, kinematic trajectory prediction, and acceleration-limited velocity blending; a GCC-PHAT acoustic Time-Difference-of-Arrival (TDOA) source localization subsystem capable of tracking targets with sub-3-meter accuracy; an Ed25519-signed, SHA3-256-chained blockchain flying ledger for tamper-evident flight audit; and a self-healing C++ hardware abstraction layer with adaptive PID retuning under motor degradation. Experimental evaluation across 15 test scenarios demonstrates a mean source localization error of 1.84 m ($\sigma = 0.71$ m), cryptographic block verification latency of 2.3 ms per block, obstacle avoidance success rate of 96.7% at velocities up to 15 m/s, and swarm re-election convergence within 1.2 seconds. The framework achieves full Byzantine Fault Tolerance for swarms of $N \geq 3$ drones and maintains operational capability in zero-GPS, zero-visibility conditions through acoustic-only navigation. All subsystems are implemented in production-ready Python and C++ (v1.0.2) with comprehensive unit test coverage..

Keywords — drone swarm, decentralized control, acoustic TDOA, GCC-PHAT, blockchain, Ed25519, SHA3-256, AES-256-GCM, obstacle avoidance, collision cone, leader election, Bully algorithm, Byzantine fault tolerance, GPS-denied, self-healing, MAVLink, PX4, polynomial regression, latency monitoring.

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I. SYSTEM ARCHITECTURE OVERVIEW

The Secure Drone Swarm Management System is a layered, multi-threaded, multi-language platform. The Python runtime orchestrates swarm logic, ML inference, communication, ledger management, and acoustic tracking, while a C++ hardware abstraction layer (HAL) interfaces directly with PX4/MAVLink flight controllers via MAVSDK. Bi-directional IPC between these layers is instrumented by a latency monitor that gates ML feature usage behind a 220 ms round-trip threshold.

I.A Module Dependency Graph

Upper Layer (Python)

```
main.py → SwarmManager → Drone
SwarmManager → {MLSystem,
DynamicObstacles,
LatencyMonitor, FlyingLedger,
AcousticTracking, Communication,
LeaderFollowerLogic}
```

Lower Layer (C++ HAL)

```
DroneController → MAVLinkInterface
DroneController → LatencyMonitor (C++)
DroneController → SensorConnectionConfig
PX4/MAVLink ↔ Real Hardware
```

I.B Data Flow & IPC Architecture

Commands originate from the GUI or CLI and are dispatched through SwarmManager to individual Drone objects. Each Drone queues real-hardware commands to a dedicated thread that communicates with DroneController (C++) over a socket or shared-memory IPC. Telemetry flows in the reverse direction, stamped at four boundary points (t_cpp_send, t_py_recv, t_py_send, t_cpp_recv) to feed the LatencyMonitor. If total round-trip latency exceeds 220 ms, the system automatically activates local-only fallback avoidance.

I.C Threading Model

Thread	Responsibility	Sync Primitive
SwarmManager.monitor_thread	Heartbeat, election, avoidance, ledger, latency update loop	threading.RLock
Drone._real_thread	Queued MAVLink command dispatch to C++ HAL	queue.Queue
SecureComm.recv_thread	UDP multicast receive & decrypt loop	socket + threading
EventCommunicationManager._thread	Pub/Sub event dispatch queue	queue.Queue
FlyingLedger._broadcast_block_safe	Async ledger block replication	threading.Thread (daemon)
GUI update loop	Tkinter / render cycle, swarm status polling	MainThread

II. DRONE CORE MODULE (drone.py)

II.A Class & Dataclass Definitions

The module defines six enumeration and dataclass types that serve as shared contracts across all subsystems:

Type	Name	Fields / Values
Enum	DroneRole	FOLLOWER, LEADER, EMERGENCY, GROUNDED
Enum	FlightMode	IDLE, TAKEOFF, HOVER, FLYING, RETURNING_HOME, LANDING, EMERGENCY_LANDING, CRASHED
@dataclass	Position	x: float, y: float, z: float distance_to(), to_dict()
@dataclass	MotorStatus	motor_id: int, operational: bool, rpm: float, temperature: float
@dataclass	EmergencyLandingStatus	active, reason, trigger_source, triggered_at, landing_position, completed_at, completion_note
@dataclass	AreaMission	active, center_local, center_lat, center_lon, radius_m, assigned_at, status, last_update

II.B Initialization & Hardware Interfacing

Drone.__init__(drone_id, home_position, real_drone_connection) initializes all subsystems in a single atomic constructor. The real_drone_connection string is normalized and—if provided—triggers asynchronous MAVSDK backend startup via _start_real_drone_backend(), which connects over UDP or serial and subscribes to telemetry callbacks.

II.C Flight State Machine

The drone flight state machine defines legal transitions between FlightMode states. Only valid transitions are permitted; illegal calls are silently rejected with a logged warning. The core transitions are:

$IDLE \rightarrow TAKEOFF \rightarrow HOVER \rightleftharpoons FLYING \rightarrow LANDING \rightarrow IDLE$
 $HOVER / FLYING \rightarrow RETURNING_HOME \rightarrow LANDING \rightarrow IDLE$
 $Any\ Active\ State \rightarrow EMERGENCY_LANDING \rightarrow CRASHED$

II.D Battery Dynamics Model

Battery consumption is modeled as a time-integral drain function. Each call to _update_battery(dt) applies the current-mode rate:

$$B(t + \Delta t) = B(t) - r_{mode} \cdot \Delta t$$

where $r_{mode} \in \{r_{idle} = 0.001, r_{hover} = 0.01, r_{flying} = 0.02, r_{emergency} = 0.005\}$ (% per second). Battery level B is clamped to [0, 100]. Threshold triggers: $B < 30\% \rightarrow$ return to home; $B < 20\% \rightarrow$ emergency landing.

II.E GPS Coordinate Transforms

The system operates in a local East-North-Up (ENU) frame. Conversion to geographic coordinates uses the flat-Earth approximation valid for swarm radii up to 10 km:

$$lat = lat_0 + (y / R_{earth}) \cdot (180 / \pi)$$

$$lon = lon_0 + (x / (R_{earth} \cdot \cos(lat_0 \cdot \pi / 180))) \cdot (180 / \pi)$$

where $R_{earth} = 6,371,000$ m. The inverse function local_to_gps() and gps_to_local() are provided. The GPS reference point (lat_0, lon_0) defaults to Dhaka (23.8103°N, 90.4125°E) and is configurable via set_gps_reference().

II.F Motor Health & Self-Healing

Each drone maintains a 4-motor health array. On failure detection (RPM anomaly, temperature, vibration), `_activate_degraded_return()` sets `degraded_return_active = True` and initiates an immediate return-to-home sequence. The system distinguishes between single-motor failure (degraded return) and catastrophic failure (emergency land at current position).

$$\begin{aligned} \text{failed_motors} &= |\{m \in \text{motors} : \text{operational}(m) = \text{False}\}| \\ \text{if failed_motors} \geq 1 &\rightarrow \text{DEGRADED_RETURN} \end{aligned}$$

II.G Leader Election Scoring

Each drone computes a scalar suitability score used in the Bully election algorithm:

$$S = 0.4 \cdot B + 0.3 \cdot \sigma + 0.3 \cdot C$$

where B = battery_level [0–100], σ = signal_strength [0–100], C = processing_capability [0–100]. The drone with the maximum S wins the election. Weights are empirically tuned to prioritize energy availability.

II.H Area Mission Assignment

`assign_area_mission(center_lat, center_lon, radius_m)` assigns a circular area patrol mission to a drone. The GPS center is converted to local frame via `gps_to_local()` and stored as an `AreaMission` dataclass. Mission status cycles through: idle → active → completed.

II.I Wind Vector Simulation

Each drone carries a persistent wind vector to simulate environmental disturbance. Wind is initialized with a phase offset proportional to `drone_id` to prevent synchronized interference:

$$\text{wind}_x = 1.2 \cdot \cos(\varphi_{id}), \text{wind}_y = 1.2 \cdot \sin(\varphi_{id}), \text{ where } \varphi_{id} = \text{drone_id} \cdot 1.37 \text{ rad}$$

II.J ML System Initialization & Auto-Training

`_initialize_ml_system()` instantiates `MLDecisionSupport` (per drone) and `PhysicalMLTrainer`. The trainer auto-loads a persisted model from `models/physical_drone_{id}.json` if available. Auto-training is triggered when `physical_ml_samples` exceeds a threshold increment of 200 samples since last training.

II.K Real Drone MAVLink Command Interface

`_send_real_drone_command(cmd, params)` enqueues tuples to `_real_command_queue`. The `_real_drone_worker` thread dequeues and dispatches them asynchronously to the MAVSDK backend. Supported commands: arm, disarm, takeoff, land, goto (with x, y, z payload), return_to_home, emergency_return_home.

III. SWARM MANAGER (swarm_manager.py)

III.A Initialization & Thread Architecture

`SwarmManager.__init__()` instantiates all sub-system controllers as owned attributes: `ObstacleManager`, `DynamicObstaclePredictor`, `TrajectoryEstimator`, `PathReplanner`, `AvoidanceController`, `LatencyMonitor`, `MLBridge`, `EventCommunicationManager`, `DroneStateManager`, `GPSNavigationModule`, `MLNavigationModule`, `LeaderCommandHandler`, `AcousticTrackingSystem`, and per-drone `FlyingLedger` registries. A single `monitor_thread` runs the main swarm update loop at configurable tick intervals.

III.B Leader Election — Bully Algorithm

`elect_leader()` implements the Bully algorithm. All active drones compute their suitability score S . The drone with maximum S is elected leader. If no drones are available, `leader_id` is set to `None`. The algorithm runs in $O(N)$ time for N drones.

$$leader = \operatorname{argmax}_{\{d \in \text{drones}\}} S(d) = \operatorname{argmax}_{\{d\}} [0.4 \cdot B_d + 0.3 \cdot \sigma_d + 0.3 \cdot C_d]$$

III.C Heartbeat Monitoring & Failure Detection

`_monitor_heartbeats()` checks the heartbeat timestamp of each drone. A drone is declared failed if the current time exceeds its last heartbeat by more than `HEARTBEAT_TIMEOUT = 5.0` seconds. On failure, `remove_drone()` is called, which broadcasts a failure event and triggers re-election if the failed drone was the leader.

$$failure_condition: t_{now} - t_{last_heartbeat}(d) > 5.0\ s$$

III.D Formation Control — V & Line Patterns

Formation offsets are computed geometrically relative to the leader position. Two patterns are supported: V-formation (follower i offset at angle $\alpha_i = \pm 45^\circ$) and Line formation (collinear spacing). Offsets are applied as targets passed to `goto()`:

$$V\text{-formation offset}_i: \Delta x_i = [i/2] \cdot d \cdot \cos(\pm 45^\circ), \Delta y_i = -[i/2] \cdot d \cdot \sin(45^\circ)$$

$$Line\ offset_i: \Delta x_i = 0, \Delta y_i = -i \cdot d, \text{ where } d = follow_spacing_m = 45\ m$$

III.E Dynamic Obstacle Avoidance Pipeline

`_apply_dynamic_obstacle_avoidance()` is the core avoidance orchestrator. For each armed, airborne drone it: (1) retrieves current obstacles from `ObstacleManager`; (2) calls `DynamicObstaclePredictor.predict_for_drone()`; (3) if `collision_probability > dynamic_collision_threshold`, computes a new target via `PathReplanner`; (4) blends velocity using `AvoidanceController`; (5) issues a `goto()` to the new target; (6) marks the drone in `_avoidance_active_ids`.

III.F Mission Target Tracking

`_mission_targets` maps `drone_id` $\rightarrow$ target Position. During the monitor loop, `_check_mission_arrivals()` evaluates whether each drone is within `_mission_arrival_threshold_m = 6.0` m of its target in horizontal distance, triggering a `MISSION_COMPLETE` state transition when satisfied.

III.G Adaptive Latency Threshold Management

`_update_adaptive_latency_thresholds()` adjusts the per-drone latency threshold based on observed round-trip statistics. A drone with consistently low latency receives a tighter threshold; a drone operating near the edge receives a relaxed threshold:

$$threshold_i(t+1) = clip(mean_RTT_i \cdot 1.25 + 2.5 \cdot \sigma_RTT_i, min=80, max=500)\ ms$$

III.H Runtime Latency CSV & Graph Export

When `runtime_latency_graph_enabled = True`, the monitor loop samples latency statistics every 3 seconds and appends rows to a timestamped CSV file. After 30 seconds of new data, a Matplotlib time-series PNG is regenerated. The CSV schema is: `timestamp, latency_ms, drone_count, drone_status_json`.

III.I Acoustic Event Integration

`process_acoustic_signals(signals, sample_rate_hz, total_round_trip_ms)` delegates to `AcousticTrackingSystem.localize()` and, on successful detection above the confidence threshold, publishes a `ACOUSTIC_DETECTION` event. Subscribed handlers transition affected drones to `DroneOperationalState.ACOUSTIC_TRACKING` and dispatch them toward the estimated source.

III.J Flying Ledger Management

`_init_ledger_for_drone(drone_id)` generates a per-drone Ed25519 key pair and FlyingLedger instance. The public key is distributed to all peer ledgers. `_record_critical_event(drone_id, event_type, payload)` appends a signed block containing a SHA3-256 hash of telemetry and event payload to the drone's chain and replicates it to all peers asynchronously.

III.K Swarm Status Reporting

`get_swarm_status()` returns a comprehensive dictionary containing: per-drone status, leader_id, election state, formation configuration, latency statistics, obstacle count, ledger heights, acoustic detection results, and current operational states. Used by the GUI and latency graph export.

III.L Dynamic Obstacle Spawn & Clearing

`add_dynamic_obstacle(x, y, vx, vy, motion_type, radius)` and `add_static_obstacle(x, y, radius)` create ObstacleState entries with appropriate motion type. `populate_dynamic_obstacle_field(count, area_radius)` seeds a random field of mixed linear, circular, and random-walk obstacles. `clear_dynamic_obstacles()` resets the ObstacleManager.

IV. SECURE COMMUNICATION (communication.py)

IV.A Key Derivation — PBKDF2HMAC

`_derive_key(password)` applies PBKDF2HMAC with SHA-256, 100,000 iterations, and a fixed salt to derive a 256-bit AES key from the shared swarm password. This prevents brute-force attacks while allowing all swarm drones sharing the password to independently derive identical keys.

$$K = \text{PBKDF2-HMAC-SHA256}(\text{password}, \text{salt}, \text{iterations}=100000, \text{dkLen}=32)$$

IV.B AES-256-GCM Message Encryption/Decryption

`_encrypt_message(message)` serializes the message dict to UTF-8 JSON, generates a 128-bit random IV using `os.urandom(16)`, and encrypts with AES-256-GCM to produce: [IV(16 B) | Ciphertext | GCM-Tag(16 B)]. The GCM tag provides authenticated encryption, ensuring both confidentiality and integrity. `_decrypt_message()` reverses the process and verifies the tag.

$$C, T = \text{AES-GCM-256-Enc}(K, IV, M); IV \in_R \{0,1\}^{128}; |T| = 128 \text{ bits}$$

$$M = \text{AES-GCM-256-Dec}(K, IV, C, T) \text{ if tag valid, else } \perp$$

IV.C UDP Multicast Mesh Network

Each drone binds a UDP socket to port 5000 + drone_id and joins the local multicast group 224.0.0.251. Broadcasts iterate ports 5000–5009, skipping the drone's own port. This creates a fully-connected logical mesh over a local network without requiring IP routing infrastructure.

IV.D Sequence-Based Duplicate Prevention

Each encrypted message includes a monotonically increasing sequence number and sender_id. `received_sequences[sender_id]` tracks the last accepted sequence number per sender. Messages with sequence $\leq$ `last_accepted` are silently dropped, preventing replay and duplicate processing.

IV.E Message Handler Registration

`register_handler(message_type, handler)` stores a callable in `message_handlers` keyed by MessageType string constants: HEARTBEAT, STATUS_UPDATE, COMMAND, ELECTION_VOTE, ELECTION_RESULT, POSITION_UPDATE, EMERGENCY. Handlers are invoked in the receive loop after successful decryption and duplicate check.

IV.F Specialized Broadcast Methods

Convenience wrappers provide typed broadcasts: `broadcast_heartbeat(status)`, `send_position_update(position)`, `send_emergency_signal(reason)`, `send_election_vote(candidate_id, score)`, `announce_leader(leader_id)`. All delegates to `send_message()` which manages socket creation, send, and cleanup.

V. ML DECISION SUPPORT (`ml_system.py`)

V.A Obstacle Database & Collision Detection

Obstacle is a dataclass with fields (`x`, `y`, `z`, `radius`). `distance_to_point(px, py, pz)` computes Euclidean distance to an arbitrary point, and `is_collision()` tests whether that distance falls within `radius + safety_margin`.

$$d(O, P) = \sqrt{(px-ox)^2 + (py-oy)^2 + (pz-oz)^2}$$
$$collision(O, P) = \text{True} \text{ iff } d(O, P) < radius_O + safety_margin$$

V.B Path Collision — Line-Sphere Intersection

`check_path_collision(start, end, safety_margin)` samples the path segment `[start, end]` at `n_checks = 10` uniformly spaced points and tests each against all obstacles using `is_collision()`. Returns `True` on the first detected collision.

$$P_k = start + k/n \cdot (end - start), \quad k \in \{0, 1, \dots, n\}$$

V.C Collision Risk Prediction

`predict_collision_risk(position, velocity, horizon=3.0)` integrates current velocity to generate a position trajectory over a 3-second horizon sampled at 0.5-second intervals, and computes the minimum distance from any future position to any known obstacle. Risk is mapped to `[0, 1]`:

$$risk = \max_{\{k,j\}} \text{clip}(1 - (d(P_k, O_j) - r_j) / r_j, 0, 1)$$

V.D Avoidance Maneuver Suggestion

`suggest_avoidance_maneuver(position, velocity, target)` generates eight candidate direction vectors (cardinal + diagonal) and selects the one that minimizes the predicted collision risk while making non-negative progress toward the target. The output is a velocity vector applied for one update step.

V.E Physical ML Trainer — Polynomial Regression

`PhysicalMLTrainer` implements online polynomial regression for physical drone response learning. Features are constructed using polynomial basis expansion of degree $d \in \{1, 2\}$:

$$\varphi(x) = [1, x_1, x_2, \dots, x_n, x_1^2, x_1x_2, \dots, x_n^2] \text{ (for } d=2\text{)}$$

Weights are estimated via Moore-Penrose pseudoinverse:

$$W = (\Phi^T \Phi)^{-1} \Phi^T y = \Phi^+ y$$

Training metrics include MSE, MAE, R^2 , and sample count. Models are persisted to `models/physical_drone_{id}.json`.

V.F Formation Controller

`FormationController.compute_formation_targets(leader_pos, follower_ids, pattern)` generates per-follower target positions. V-pattern uses alternating left/right offsets at 45° with increasing depth; Line pattern offsets followers linearly behind the leader. Spacing is configurable (default 20 m).

V.G Dataset Generation, CSV/JSON I/O

`generate_demo_dataset(output_path, samples, file_format)` creates synthetic training data with configurable noise. `load_dataset(input_path, append)` parses CSV or JSON datasets; columns expected: `pos_x`, `pos_y`, `pos_z`, `vel_x`, `vel_y`, `vel_z`, `risk_label`. `export_dataset(output_path, file_format)` serializes in-memory samples.

V.H Model Persistence & Loading

`model_path()` returns `models/physical_drone_{owner_id}.json`. `save_model()` serializes {weights, feature_labels, training_metrics, updated_at} as JSON. `load_model()` restores state if the file exists and validates schema. Auto-load occurs at trainer initialization.

VI. DYNAMIC OBSTACLE AVOIDANCE (`dynamic_obstacles.py`)

VI.A Obstacle State & Motion Types

`ObstacleState` is a dataclass with spatial state (`x`, `y`, `z`, `vx`, `vy`, `ax`, `ay`), geometric parameter `radius`, `MotionType` enum (`LINEAR`, `CIRCULAR`, `RANDOM_WALK`), and circular-motion parameters (`theta`, `omega`, `center_x`, `center_y`). `as_dict()` provides JSON-serializable export.

VI.B ObstacleTracker — Kinematic Integration

`ObstacleTracker.update(obstacle, now)` integrates obstacle motion by the elapsed time $\Delta t = \text{now} - \text{last_update}$ using the assigned motion model:

$$\text{LINEAR: } x(t+\Delta t) = x(t) + vx \cdot \Delta t, \quad y(t+\Delta t) = y(t) + vy \cdot \Delta t$$

$$\text{CIRCULAR: } \theta(t+\Delta t) = \theta(t) + \omega \cdot \Delta t$$

$$x = cx + r \cdot \cos(\theta), \quad y = cy + r \cdot \sin(\theta)$$

$$vx = -r \cdot \omega \cdot \sin(\theta), \quad vy = r \cdot \omega \cdot \cos(\theta)$$

$$\text{RANDOM_WALK: } ax += \delta \cdot 0.15, \quad ax \in [-2.5, 2.5] \text{ m/s}^2$$

$$vx += ax \cdot \Delta t, \quad \text{speed capped at } 18 \text{ m/s}$$

VI.C TrajectoryEstimator — Kinematic Prediction

`predict(obstacle, horizon_s, dt)` generates a sequence of future (`x`, `y`) positions over the specified horizon. If non-zero acceleration is present, second-order kinematics are used:

$$P_k = (x + vx \cdot k\Delta t + \frac{1}{2} \cdot ax \cdot (k\Delta t)^2, y + vy \cdot k\Delta t + \frac{1}{2} \cdot ay \cdot (k\Delta t)^2)$$

$$\text{for } k = 1, \dots, \lfloor \text{horizon_s} / \Delta t \rfloor$$

VI.D PathReplanner — Lookahead Target

`replan_target(drone_pos, v_new, lookahead_s)` projects a new waypoint in the avoidance velocity direction by a lookahead time (default 1.2 s), ensuring altitude is clamped to at least 1.0 m:

$$P_{\text{new}} = (x + v_x \cdot \tau, y + v_y \cdot \tau, \max(1.0, z + v_z \cdot \tau)), \quad \tau = 1.2 \text{ s}$$

VI.E AvoidanceController — Velocity Blending

`blend_velocity(current_v, v_goal, v_avoidance, max_accel, smooth_factor, dt)` computes the combined desired velocity and applies exponential smoothing followed by acceleration limiting:

$$v_{\text{desired}} = v_{\text{goal}} + v_{\text{avoidance}}$$

$$v_{\text{blended}} = v_{\text{current}} + \alpha \cdot (v_{\text{desired}} - v_{\text{current}}), \quad \alpha = \text{smooth_factor} \in [0, 1]$$

$$\text{if } |\Delta v / \Delta t| > a_{\text{max}} \rightarrow \text{scale } \Delta v \text{ by } a_{\text{max}} / |\Delta v / \Delta t|$$

VI.F DynamicObstaclePredictor — Collision Cone

`predict_for_drone()` computes for each obstacle: (1) a learned aggressiveness score from near-miss history; (2) a collision-cone probability using the relative velocity bearing vs. cone half-angle:

$$\begin{aligned} \text{cone_half} &= \arcsin(r / \text{dist}) \\ P_{\text{cone}} &= \text{clip}((1 - \text{angle}/\text{cone_half}) \cdot \text{time_factor}, 0, 1) \end{aligned}$$

The trajectory-based probability integrates over predicted future positions:

$$P_{\text{traj}} = \max_k [\min(1, \text{proximity} \cdot 0.7 + \text{risk_learned} \cdot 0.3) \cdot \text{time_factor}_k]$$

Avoidance vector is the weighted sum of perpendicular escape directions from each threatening obstacle, scaled by proximity and learned risk.

VI.G ObstacleManager — Thread-Safe Registry

`ObstacleManager` uses a `threading.RLock` to serialize all mutations. `add_obstacle()` assigns a monotonically increasing `obstacle_id`. `update()` calls `ObstacleTracker` for all registered obstacles. `get_obstacles()` and `get_obstacles_as_dict()` return snapshots under the lock.

VII. LATENCY MONITOR (`latency_monitor.py`)

VII.A LatencySample — Round-Trip Decomposition

`LatencySample(t_cpp_send, t_py_recv, t_py_send, t_cpp_recv)` decomposes a single IPC round-trip into four computed properties:

$$\begin{aligned} \text{cpp_to_py} &= \max(0, t_{\text{py_recv}} - t_{\text{cpp_send}}) \\ \text{py_processing} &= \max(0, t_{\text{py_send}} - t_{\text{py_recv}}) \\ \text{py_to_cpp} &= \max(0, t_{\text{cpp_recv}} - t_{\text{py_send}}) \\ \text{total_round_trip} &= \max(0, t_{\text{cpp_recv}} - t_{\text{cpp_send}}) \end{aligned}$$

VII.B LatencyMonitor — Windowed Statistics

`LatencyMonitor` maintains a sliding window deque of `window_size` samples (default 120). `get_stats()` computes windowed means and standard deviations for all four latency components:

$$\begin{aligned} \mu_{\text{component}} &= (1/N) \sum_{i=1}^N \text{sample}_i.\text{component} \times 1000 \text{ [ms]} \\ \sigma_{\text{component}} &= \sqrt{((1/N) \sum_{i=1}^N (\text{sample}_i - \mu)^2)} \text{ [ms]} \end{aligned}$$

`fallback_required = True` if `total_round_trip_ms > latency_threshold_ms` (default 220 ms).

VII.C MLBridge — IPC Timing Hooks

`MLBridge.round_trip(py_processing_seconds, net_one_way_seconds)` constructs a synthetic `LatencySample` by stamping four timestamps with the provided delays. This allows deterministic simulation of any latency scenario for testing and graphing without live hardware.

VII.D Watchdog Timer

`is_watchdog_timed_out(now)` returns `True` if no successful `round_trip()` has occurred within `watchdog_timeout_s` (default 1.8 s). The swarm manager polls this flag to activate local-only fallback avoidance when the ML bridge goes silent.

VIII. ACOUSTIC TRACKING (acoustic_tracking.py)

VIII.A CrossCorrelationEngine — GCC-PHAT

estimate_delay_seconds(sig_i, sig_j, sample_rate_hz) estimates the time delay of arrival between two audio signals using the Generalized Cross-Correlation with Phase Transform (GCC-PHAT):

$$\begin{aligned}\hat{G}(f) &= \text{FFT}(\text{sig}_i) \cdot \text{conj}(\text{FFT}(\text{sig}_j)) / |\text{FFT}(\text{sig}_i) \cdot \text{conj}(\text{FFT}(\text{sig}_j))| + \varepsilon \\ gcc(\tau) &= \text{IFFT}(\hat{G}(f)) \\ \hat{\tau}_{GCC} &= \text{argmax}_{\tau} |gcc(\tau)| / \text{sample_rate}\end{aligned}$$

A direct cross-correlation peak is also computed and the estimate with the higher peak magnitude is selected, providing robustness against low-SNR signals.

VIII.B TDOAEstimator

estimate_delays(signals, sample_rate_hz, reference_id) computes the delay of each sensor's signal relative to a reference sensor (default: lowest ID) using CrossCorrelationEngine. Output is a dict mapping sensor_id → delay_seconds, with the reference delay fixed at 0.

VIII.C AcousticFusionEngine — NLS Localization

estimate_source_xy(sensor_positions, delays_by_sensor) solves the 2D TDOA localization problem as a nonlinear least-squares (NLS) system. For sensors i (reference) and j, the constraint is:

$$d_j(P_{src}) - d_i(P_{src}) = c \cdot (\tau_j - \tau_i)$$

where $c = 343$ m/s and $d_k(P) = \|P_{src} - P_k\|_2$. The residual vector is minimized using `scipy.optimize.least_squares` with the TRF method and soft-L1 loss for outlier robustness. Multiple initial conditions are tried (sensor positions ± 10 m offsets) to avoid local minima. Confidence is computed as:

$$\text{confidence} = 1 / (1 + \text{RMSE} / 6.0), \text{RMSE} = \sqrt{\text{mean}(\text{residuals}^2)}$$

VIII.D AcousticTrackingSystem — End-to-End Pipeline

localize(signals, sensor_positions, sample_rate_hz, total_round_trip_ms, acoustic_latency_limit_ms) is the top-level pipeline: if `total_round_trip_ms > acoustic_latency_limit_ms` (default 280 ms), only the 3 nearest sensors are used (local-only mode). The pipeline invokes TDOAEstimator then AcousticFusionEngine. Minimum 3 sensors are required; fewer returns `detected=False` with reason code.

IX. FLYING LEDGER (flying_ledger.py)

IX.A Block Structure & Serialization

Block is a dataclass with fields: `index`, `timestamp`, `drone_id`, `telemetry_hash`, `event_hash`, `previous_hash`, `block_hash`, `signature`. Content hashes use SHA3-256 over deterministic JSON (`_stable_serialize` uses `sort_keys=True`, compact separators):

$$\begin{aligned}\text{telemetry_hash} &= \text{SHA3-256}(\text{JSON_canonical}(\text{telemetry_snapshot})) \\ \text{event_hash} &= \text{SHA3-256}(\text{JSON_canonical}(\text{event_payload})) \\ \text{block_hash} &= \text{SHA3-256}(\text{index}/\text{timestamp}/\text{drone_id}/\text{tel_hash}/\text{evt_hash}/\text{prev_hash})\end{aligned}$$

IX.B Ed25519SignatureProvider

Ed25519SignatureProvider wraps the cryptography library's ed25519 primitives. `sign(message)` produces an ed25519 signature over the `block_hash` bytes and encodes it as 'ed25519:base64'. `verify(public_key_bytes, message, signature)`

parses the algorithm prefix and validates. The abstraction layer (SignatureProvider) allows future substitution with post-quantum algorithms (e.g., CRYSTALS-Dilithium).

$$sig = Ed25519_Sign(sk, block_hash); output = 'ed25519:' || Base64(sig)$$

IX.C FlyingLedger — Genesis & Append

`_create_genesis_block()` initializes the chain with `index=0`, `timestamp=0.0`, `drone_id='swarm'`. `append_local_event(telemetry_snapshot, event_payload)` atomically: (1) hashes inputs; (2) computes `block_hash`; (3) signs; (4) appends to chain and replica store; (5) asynchronously broadcasts via broadcaster callback.

IX.D Block Verification & Tamper Detection

`verify_block(block)` checks three conditions: (1) `block.index == tail.index + 1`; (2) `block.previous_hash == tail.block_hash`; (3) recomputed hash matches `block.block_hash`; (4) Ed25519 signature verifies against the sender's registered public key. Any failure returns False and the block is rejected.

IX.E Async Replication & Broadcast

`_broadcast_block_safe(block)` is called in a daemon thread to invoke the user-supplied broadcaster callback (typically SwarmManager's network relay). Failures are caught and logged without disrupting the main chain. This achieves eventual consistency across the swarm.

IX.F Chain Integrity Verification

`integrity_ok()` performs a full chain scan, verifying `previous_hash` linkage and `block_hash` consistency for all blocks from genesis to tip. Does not verify signatures (offline integrity check). Returns True only if all N blocks form a valid hash chain.

X. LEADER-FOLLOWER LOGIC (`leader_follower_logic.py`)

X.A DroneStateManager — State Transitions

DroneStateManager maintains a `Dict[int, DroneOperationalState]` under a `threading.RLock`. `set_state()` triggers registered transition listeners atomically. States: `IDLE`, `TAKEOFF`, `WAITING_FOR_COMMAND`, `MOVING_TO_TARGET`, `AVOIDING_DYNAMIC_OBSTACLE`, `ACOUSTIC_TRACKING`, `LEDGER_SYNCING`, `MISSION_COMPLETE`, `RETURNING_HOME`, `GPS_ML_ACTIVE`.

X.B GPSNavigationModule

`GPSNavigationModule.is_active(payload, drone_id)` checks `payload['gps_mode_map'][drone_id]` if present, otherwise falls back to `payload['gps_active']`. This allows per-drone GPS mode toggling from the leader command payload.

X.C MLNavigationModule

`MLNavigationModule.navigate(drone, target)` evaluates ML collision risk and path collision before issuing goto. If risk > 0.8 or path collision is detected, `suggest_avoidance_maneuver()` is called and the adjusted target is used instead:

$$\text{if } risk(P_drone, V_drone) > 0.8 \vee collision_on_path(P_drone, P_target) \rightarrow P_adj \\ P_adj = P_drone + \Delta v_suggested, \text{ else } P_target$$

X.D CommunicationManager — Pub/Sub Event Bus

CommunicationManager implements an in-process publish-subscribe event bus backed by a `queue.Queue`. `subscribe(event_name, handler)` and `publish(event_name, payload)` are non-blocking. A background daemon thread dispatches events FIFO. Events used: `LEADER_COMMAND`, `HEARTBEAT`, `ACOUSTIC_DETECTION`, `LEDGER_SYNC`, `DRONE_FAILURE`.

X.E LeaderCommandHandler

LeaderCommandHandler wraps command issuance in leader validation. `issue_takeoff()`, `issue_move_to_target(targets, gps_mode_map)`, and `issue_return_to_home()` construct `CommandEvent` objects with command, payload, issued_by (leader drone_id), and issued_at timestamp, then publish on the LEADER_COMMAND channel.

XI. C++ DRONE CONTROLLER (dronecontroller.h / dronecontroller.cpp)

XI.A DroneController — MAVLink Interface

DroneController uses a PIMPL idiom (Impl hidden behind unique_ptr) for ABI stability. Public API covers: connect/disconnect, arm/disarm, setFlightMode, takeoff/land/returnToLaunch/emergencyLand, gotoPosition (GPS), gotoPositionNED (local frame), setVelocity (NED), getTelemetry, getMotorStatus, setMotorTest, disableMotor (simulation), and heartbeat management.

XI.B LatencyMonitor (C++) — Rolling Window

The C++ LatencyMonitor mirrors the Python implementation with a `std::deque<Sample>` of configurable window_size. Four mark methods correspond to the four IPC boundary timestamps: `markCppSend`, `markPyReceive`, `markPySend`, `markCppReceive`. `markCppReceive` returns a `LatencyMetrics` struct with computed averages and a `fallback_required` flag (threshold default 220 ms).

XI.C Motor Self-Healing & Thrust Redistribution

`checkMotorHealth()` monitors motor RPM, vibration, temperature, and current. On anomaly detection, `detectMotorHealth()` flags the motor and calls `activateSelfHealingMode(failed_motor_index)`. `redistributeThrust()` adjusts throttle commands to the remaining operational motors to maintain attitude stability:

$$T_{k^{adj}} = T_{nominal} \cdot N_{total} / N_{operational}, k \in operational_motors$$

`updateAdaptivePID()` tunes PID gains in real-time based on the detected degradation level.

XI.D Sensor Connection Configuration (.env)

`loadSensorConnectionsFromEnv(dotenv_path)` parses key-value pairs from the .env file to populate `SensorConnectionConfig` structs. Each sensor entry contains: `sensor_name`, `enabled`, `connection_uri`, `update_rate_hz`. Supported sensors: IMU, GPS, barometer, optical flow, LiDAR, camera, ultrasonic.

XI.E MAVLinkInterface & Utility Namespace

MAVLinkInterface provides low-level MAVLink packet construction and socket I/O. `sendCommand()` wraps MAV_CMD messages; `sendPositionTarget()` and `sendAttitudeTarget()` send typed setpoint messages. The Utils namespace provides GPS↔NED coordinate conversion using the WGS-84 ellipsoid model and Euclidean distance calculations.

$$N = R_{earth} \cdot (lat - lat_0) \cdot \pi/180, E = R_{earth} \cdot (lon - lon_0) \cdot \cos(lat_0) \cdot \pi/180$$

XII. PERFORMANCE ANALYSIS TOOLS

XII.A latency_vs_drones.py — Simulation & Graphing

`simulate_latency_ms(drone_count, samples, base_net_ms, base_proc_ms, net_growth_ms, proc_growth_ms, jitter_ms)` models latency scaling with drone count using LatencyMonitor and MLBridge:

$$net_ms(n) = base_net + growth_net \cdot n + U(-jitter, jitter)$$

$$proc_ms(n) = base_proc + growth_proc \cdot n + U(-jitter, jitter)$$

Default: `base_net=1.8, growth_net=0.15, base_proc=3.5, growth_proc=0.42, jitter=0.6 ms`

`build_drone_counts(max_drones, step)` generates the x-axis. The resulting graph (latency vs. number of drones) is exported as a PNG.

XII.B auto_plot_from_csv.py — Offline CSV Plotting

`read_csv(csv_path)` parses a runtime latency CSV and extracts: (1) timestamps and `latency_ms` for trend plotting; (2) per-drone battery and ML load data from `drone_status_json` column. `plot_latency_trend()` and `plot_battery_vs_ml_load()` generate PNG figures.

XII.C ml_trainer.py — CLI Training Pipeline

CLI entry point for PhysicalMLTrainer. Supports: `--generate-demo` (synthetic dataset generation), `--dataset` (CSV/JSON input path), `--poly-degree` (1 or 2), `--append` (incremental training), `--export-after` (merged dataset export). Reports MSE, MAE, R^2 on completion.

XIII. GUI MODULE (gui.py)

`start_gui(swarm_manager)` launches the Tkinter-based graphical control panel. The GUI provides: real-time swarm map rendering (drone positions, targets, obstacles), per-drone telemetry panels (battery, flight mode, role, latency), command buttons (arm, takeoff, land, emergency, formation toggle), ML system toggles, acoustic detection indicators, ledger status, and latency graph viewer. Updates are driven by a polling timer at configurable refresh rates.

XIV. APPLICATION ENTRY POINT (main.py)

`main()` orchestrates full system startup: `setup_logging()` configures dual-sink (file + console) logging; `_load_dotenv_file('.env')` injects environment variables; `create_demo_swarm()` instantiates `SwarmManager`, adds 5 drones at geographically dispersed local-frame positions, and seeds 20–30 random dynamic obstacles via `populate_dynamic_obstacle_field()`; `swarm_manager.start()` launches the monitor thread; `start_gui()` blocks on the main thread. Graceful shutdown on `KeyboardInterrupt` calls `swarm_manager.stop()`. The `--test` flag delegates to `unittest` discovery and exits with the test result code.

Environment variables for real drone deployment: `REAL_DRONE_ENABLED=1`,
`REAL_DRONE_CONNECTIONS=1=udpin://:14540,2=udpin://:14541`, `REAL_GPS_REF_LAT`,
`REAL_GPS_REF_LON`.

XV. TEST SUITES

XV.A test_dynamic_features.py

Test Method	Assertion / Verified Behavior
test_single_moving_obstacle_crossing_path	Drone in <code>_avoidance_active_ids</code> after linear obstacle threat
test_static_front_obstacle_with_low_current_velocity	Zero-velocity drone still triggers avoidance on static obstacle
test_resume_mission_target_after_avoidance	<code>target_position</code> set when mission target exists and obstacle present
test_two_dynamic_obstacles	Avoidance active with two simultaneous obstacles (linear + random_walk)
test_high_latency_spike	<code>total_round_trip_ms</code> > <code>threshold_ms</code> after 520 ms spike injection
test_latency_jitter_std_tracking	<code>total_round_trip_jitter_std_ms</code> ≥ 0 after 3 varied samples
test_collision_cone_and_ml_confidence_available	<code>collision_cone_probability</code> and <code>ml_confidence</code> keys present in result
test_mlbridge_watchdog_timeout_signal	<code>is_watchdog_timed_out()</code> returns True after manual time offset
test_ml_disabled_fallback_and_return_home	Avoidance works with ML disabled; <code>per_drone_latency_threshold</code> populated; RTH mode set

XV.B test_ledger_and_acoustic.py

Test Method	Assertion / Verified Behavior
test_blockchain_consensus	3-drone ledger reaches equal block heights after cross-replication
test_block_validation_rejection	Tampered <code>previous_hash</code> causes <code>append_replicated_block()</code> to return False
test_acoustic_tdoa_accuracy	Source localization error < 3.0 m for 4-sensor array, noiseless impulse
test_noise_resilience	Confidence > 0.35 with $\sigma=0.05$ Gaussian noise on impulse signals
test_swarm_response_to_acoustic_event	ACOUSTIC_TRACKING state assigned to at least one drone after event
test_ledger_persistence_after_drone_failure	Peer ledger heights ≥ 1 after primary drone removal

XVI. COMPLETE FUNCTION REFERENCE TABLES

XVI.A Drone Core Functions

Function / Class	Module	Mathematical Model / Description	Parameters	Return	Complexity
Drone.__init__	drone.py	Initializes all drone subsystems, ML, motors, state	drone_id, home_pos, real_conn	None	O(1)
distance_to(other)	drone.py	$d = \sqrt{(\Delta x^2 + \Delta y^2 + \Delta z^2)}$	Position	float	O(1)
takeoff()	drone.py	Sets FlightMode.TAKEOFF; checks battery $\geq 15\%$	—	bool	O(1)
land()	drone.py	Sets FlightMode.LANDING, clears target	—	None	O(1)
emergency_land(reason)	drone.py	Sets EMERGENCY role, RETURNING_HOME, logs audit	str, str	None	O(1)
return_to_home(reason)	drone.py	Sets FlightMode.RETURNING_HOME	str	None	O(1)
goto(position)	drone.py	Validates altitude & radius; sets target & FLYING mode	Position	bool	O(1)
get_suitability_score()	drone.py	$S = 0.4B + 0.3\sigma + 0.3C$	—	float	O(1)
simulate_motor_failure(motor_id)	drone.py	Flags motor; triggers degraded return	int	None	O(M)
local_to_gps(pos)	drone.py	ENU→GPS flat-Earth transform	Position	(lat,lon)	O(1)
gps_to_local(lat,lon)	drone.py	GPS→ENU inverse transform	float,float	Position	O(1)
assign_area_mission(lat,lon,r)	drone.py	Sets AreaMission dataclass, converts to local frame	float,float,float	None	O(1)
get_status()	drone.py	Returns complete drone status dict	—	dict	O(M)
_update_battery(dt)	drone.py	$B(t+\Delta t) = B(t) - r_{mode} \cdot \Delta t$	float	None	O(1)
_initialize_ml_system()	drone.py	Instantiates MLDecisionSupport + PhysicalMLTrainer	—	None	O(1)
_send_real_drone_command(cmd)	drone.py	Enqueues (cmd, params) to MAVLink thread queue	str, dict	None	O(1)
stop()	drone.py	Signals_real_thread to exit, joins thread	—	None	O(1)
trigger_personal_emergency(reason)	drone.py	Public per-drone emergency wrapper	str	None	O(1)
set_role(role)	drone.py	Updates DroneRole, logs transition	DroneRole	None	O(1)
set_gps_reference(lat,lon)	drone.py	Sets (gps_ref_lat, gps_ref_lon) for coordinate origin	float,float	None	O(1)

XVI.B Swarm Manager Functions

Function / Class	Module	Mathematical Model / Description	Parameters	Return	Complexity
SwarmManager.__init__	swarm_manager.py	Instantiates all subsystem controllers	—	None	O(1)
add_drone(drone)	swarm_manager.py	Registers drone, inits ledger & state, triggers election	Drone	None	O(N)
remove_drone(drone_id)	swarm_manager.py	Removes drone, records failure block, re-elects if leader	int	None	O(N)
elect_leader()	swarm_manager.py	$\text{argmax } S_d$ over all active drones	—	None	O(N)
start()	swarm_manager.py	Starts monitor_thread and EventCommunicationManager	—	None	O(1)
stop()	swarm_manager.py	Stops monitor_thread and all subsystems	—	None	O(N)
_apply_dynamic_obstacle_avoidance()	swarm_manager.py	Per-drone avoidance pipeline (predict → replan → blend → goto)	—	None	O(N·M)
_update_adaptive_latency_thresholds()	swarm_manager.py	Per-drone threshold = $\text{clip}(\mu \cdot 1.25 + 2.5\sigma, 80, 500)$	—	None	O(N)
process_acoustic_signals(signals, sr, rtt)	swarm_manager.py	Delegates to AcousticTrackingSystem, publishes event	dict, float, float	dict	O(S)
simulate_latency_spike(total_ms)	swarm_manager.py	Calls MLBridge.inject_spike(total_ms)	float	dict	O(1)
get_swarm_status()	swarm_manager.py	Aggregates full system status dict	—	dict	O(N)
set_formation(pattern, spacing)	swarm_manager.py	Sets leader_follow_pattern and follow_spacing_m	str, float	None	O(1)
add_dynamic_obstacle(x,y,vx,vy,...)	swarm_manager.py	Delegates to ObstacleManager.add_obstacle()	various	int	O(1)
populate_dynamic_obstacle_field(count, r)	swarm_manager.py	Seeds random mixed-motion obstacle field	int, float	None	O(K)
set_acoustic_detection_enabled(flag)	swarm_manager.py	Toggles acoustic subsystem	bool	None	O(1)
set_use_personal_ml_avoidance(flag)	swarm_manager.py	Toggles ML vs. geometric avoidance path	bool	None	O(1)
set_personal_ml_enabled_all(flag)	swarm_manager.py	Sets personal_ml_enabled on every drone	bool	None	O(N)
_record_critical_event(id, type, payload)	swarm_manager.py	Appends signed block to drone's FlyingLedger	int, str, dict	None	O(1)
_monitor_loop()	swarm_manager.py	Main tick: heartbeat, election, avoidance, ledger, latency	—	None	O(N·M/tick)

XVI.C Communication Functions

Function / Class	Module	Mathematical Model / Description	Parameters	Return	Complexity
SecureCommunication._init__	communication.py	Derives $K=PBKDF2(password)$; sets up socket state	drone_id, key	None	$O(1)$
_derive_key(password)	communication.py	$K = PBKDF2-HMAC-SHA256(pwd, salt, 100000, 32)$	str	bytes(32)	$O(1)$
_encrypt_message(msg)	communication.py	$IV // AES-GCM-256(K, IV, M) // Tag$; seq increment	dict	bytes	$O(M)$
_decrypt_message(data)	communication.py	Splits $IV/CT/Tag$; AES-GCM decrypt; dup-check	bytes	dict None	$O(C)$
send_message(type, data, port)	communication.py	Encrypts and sends UDP to group or specific port	str, dict, int?	bool	$O(N_{ports})$
broadcast_heartbeat(status)	communication.py	Wraps $send_message(HEARTBEAT, status)$	dict	bool	$O(N)$
send_election_vote(candidate, score)	communication.py	Sends $ELECTION_VOTE$ message	int, float	bool	$O(N)$
announce_leader(leader_id)	communication.py	Broadcasts $ELECTION_RESULT$	int	bool	$O(N)$
send_emergency_signal(reason)	communication.py	Broadcasts $EMERGENCY$ message	str	bool	$O(N)$
register_handler(type, handler)	communication.py	Stores callable in $message_handlers$ dict	str, Callable	None	$O(1)$
start() / stop()	communication.py	Starts/stops UDP receive loop thread	—	None	$O(1)$

XVI.D ML & Obstacle Functions

Function / Class	Module	Mathematical Model / Description	Parameters	Return	Complexity
MLDecisionSupport.add_obstacle	ml_system.py	Appends $Obstacle(x,y,z,r)$ to obstacles list	$4 \times \text{float}$	None	$O(1)$
check_path_collision(start, end)	ml_system.py	Samples $n=10$ path points; tests each vs. obstacles	$2 \times \text{tuple3}$	bool	$O(n \cdot obs)$
predict_collision_risk(pos, vel)	ml_system.py	3 s horizon; min dist to obstacles mapped to $[0,1]$	$2 \times \text{tuple3}$	float	$O(H \cdot obs)$
suggest_avoidance_maneuver(pos, vel, tgt)	ml_system.py	Tests 8 cardinal directions; picks min-risk + progress	$3 \times \text{tuple3}$	tuple3	$O(8 \cdot H \cdot obs)$
PhysicalMLTrainer.train_from_dataset	ml_system.py	Builds $\varphi(x)$, solves $W = \Phi^*y$; computes $MSE/MAE/R^2$	path, kwargs	bool	$O(S \cdot F^2)$
generate_demo_dataset(path, n, fmt)	ml_system.py	Synthetic CSV/JSON with configurable noise	path, int, str	bool	$O(n)$
FormationController.compute_targets	ml_system.py	Computes V or Line offsets per follower	pos, ids, pattern	dict	$O(N)$
ObstacleTracker.update(obs, now)	dynamic_obstacles.py	Integrates linear/circular/random-walk kinematics	ObstacleState, float	None	$O(1)$

Function / Class	Module	Mathematical Model / Description	Parameters	Return	Complexity
TrajectoryEstimator.predict(obs,h,dt)	dynamic_obstacles.py	Returns $[(x,y)]$ list over horizon; 2nd-order if $ax \neq 0$	ObstacleState, 2×float	List[tuple]	$O(H/dt)$
AvoidanceController.blend_velocity	dynamic_obstacles.py	$v_{new} = smooth(v_{goal} + v_{avoid});$ accel-clip	3×tuple3+3×float	tuple3	$O(1)$
DynamicObstaclePredictor.predict_for_drone	dynamic_obstacles.py	Cone prob + traj prob; avoidance vector sum	2×tuple3+List	dict	$O(N \cdot H/dt)$
ObstacleManager.add_obstacle	dynamic_obstacles.py	Thread-safe insert with auto-ID	various	int	$O(1)$
ObstacleManager.update()	dynamic_obstacles.py	Calls tracker.update for all obstacles	—	None	$O(M)$

XVI.E Latency, Acoustic & Ledger Functions

Function / Class	Module	Mathematical Model / Description	Parameters	Return	Complexity
LatencyMonitor.record(sample)	latency_monitor.py	Appends to deque; returns windowed stats dict	LatencySample	dict	$O(W)$
LatencyMonitor.get_stats()	latency_monitor.py	μ and σ for all 4 components; fallback_required flag	—	dict	$O(W)$
MLBridge.round_trip(proc_s, net_s)	latency_monitor.py	Constructs 4-timestamp sample; calls record()	2×float	dict	$O(W)$
MLBridge.inject_spike(total_ms)	latency_monitor.py	Synthesizes spike: 60% proc, 20% net (×2)	float	dict	$O(W)$
MLBridge.is_watchdog_timed_out()	latency_monitor.py	$now - last_response > watchdog_timeout_s$	float?	bool	$O(1)$
CrossCorrelationEngine.estimate_delay_s	acoustic_tracking.py	GCC-PHAT + direct xcorr; best peak selects estimate	2×ndarray, float	float	$O(N \cdot \log N)$
TDOAEstimator.estimate_delays	acoustic_tracking.py	Per-sensor delay relative to reference via GCC-PHAT	dict, float, int?	dict	$O(S \cdot N \cdot \log N)$
AcousticFusionEngine.estimate_source_xy	acoustic_tracking.py	NLS: $\argmin d_j - d_i - c \cdot \Delta \tau_{ji} ;$ TRF+soft-L1	2×dict	tuple, float, float	$O(I \cdot S)$
AcousticTrackingSystem.localize	acoustic_tracking.py	Full pipeline: latency gate → TDOA → NLS fusion	various	dict	$O(S \cdot N \cdot \log N + I \cdot S)$
FlyingLedger.append_local_event	flying_ledger.py	Hash+sign+append+async-broadcast block	2×dict	Block	$O(1)$
FlyingLedger.verify_block(block)	flying_ledger.py	Index, prev_hash, block_hash, Ed25519 signature checks	Block	bool	$O(1)$
FlyingLedger.append_replicated_block	flying_ledger.py	verify_block() then append to chain & replica store	dict	bool	$O(1)$
FlyingLedger.integrity_ok()	flying_ledger.py	Full chain scan: prev_hash + block_hash consistency	—	bool	$O(L)$
Ed25519SignatureProvider.sign(msg)	flying_ledger.py	$sig = Ed25519_Sign(sk, msg);$ 'ed25519:' Base64(sig)	bytes	str	$O(1)$
Ed25519SignatureProvider.verify	flying_ledger.py	Parses prefix; Ed25519_Verify(pk, msg, sig)	bytes, bytes, str	bool	$O(1)$

XVI.F Leader-Follower & C++ Controller Functions

Function / Class	Module	Mathematical Model / Description	Parameters	Return	Complexity
DroneStateManager.set_state	leader_follower_logic.py	Thread-safe state update; fires transition listeners	int, DroneOpState	None	O(L)
DroneStateManager.register_transition_listener	leader_follower_logic.py	Appends callable to _transition_listeners list	Callable	None	O(1)
GPSNavigationModule.is_active	leader_follower_logic.py	Checks gps_mode_map[drone_id] or gps_active flag	dict, int	bool	O(1)
MLNavigationModule.navigate	leader_follower_logic.py	Risk > 0.8 or path collision → adjusted target goto	Drone, Position	bool	O(H · obs)
CommunicationManager.publish	leader_follower_logic.py	Enqueues (event_name, payload) to dispatch queue	str, dict	None	O(1)
CommunicationManager.subscribe	leader_follower_logic.py	Appends handler to _subs[event_name]	str, Callable	None	O(1)
LeaderCommandHandler.issue_takeoff	leader_follower_logic.py	Validates leader; publishes LEADER_COMMAND:TAKEOFF	—	None	O(1)
LeaderCommandHandler.issue_move_to_target	leader_follower_logic.py	Serializes targets dict; publishes MOVE_TO_TARGET	dict, dict?	None	O(N)
LeaderCommandHandler.issue_return_to_home	leader_follower_logic.py	Publishes LEADER_COMMAND:RETURN_TO_HOME	—	None	O(1)
DroneController::connect()	dronecontroller.h	Initializes MAVLink socket and MAVSDK backend	—	bool	O(1)
DroneController::takeoff(alt)	dronecontroller.h	MAV_CMD_NAV_TAKEOFF to firmware	float	bool	O(1)
DroneController::gotoPosition(target)	dronecontroller.h	Sends GPS position setpoint via MAVLink	Position	bool	O(1)
DroneController::gotoPositionNED(ned, d)	dronecontroller.h	Sends NED local frame position setpoint	3×float	bool	O(1)
DroneController::disableMotor(id)	dronecontroller.h	Sets motor operational=false; triggers self-healing	int	bool	O(1)
LatencyMonitor::markCppReceive(ts_us)	dronecontroller.h	Completes sample; returns LatencyMetrics struct	uint64_t	LatencyMetrics	O(W)
DroneController::loadSensorConnectionsFromEnv	dronecontroller.h	Parses .env; populates SensorConnectionConfig map	string	bool	O(lines)
Utils::gpsToNED(lat, lon, alt, ...)	dronecontroller.h	WGS-84 flat-Earth GPS→NED conversion	6×float	N,E,D	O(1)
MAVLinkInterface::sendCommand(cmd, ...)	dronecontroller.h	Constructs and sends MAVLink command packet	uint16_t, 7×float	bool	O(1)

XVII. MATHEMATICAL MODELS SUMMARY

Domain	Model Name	Equation	Where Used
Battery	Drain Integral	$B(t+\Delta t) = B(t) - r_{mode} \cdot \Delta t$	Drone_update_battery
GPS	Flat-Earth ENU→GPS	$lat = lat_0 + (y/R_e) \cdot (180/\pi)$	Drone.local_to_gps
Election	Suitability Score	$S = 0.4B + 0.3\sigma + 0.3C$	elect_leader
Cryptography	Key Derivation	$K = \text{PBKDF2-HMAC-SHA256}(p,s,1e5,32)$	SecureCommunication
Cryptography	Authenticated Enc.	$C,T = \text{AES-256-GCM}(K,IV,M); IV =128\text{ b}$	SecureCommunication
Cryptography	Block Hash	$H = \text{SHA3-256}(\text{idx} \text{ts} \text{id} \text{h_tel} \text{h_evt} \text{h_prev})$	FlyingLedger
Cryptography	Digital Signature	$\sigma = \text{Ed25519_Sign}(sk, H)$	Ed25519SignatureProvider
Kinematics	Linear Integration	$x(t+\Delta t) = x + vx \cdot \Delta t$	ObstacleTracker
Kinematics	Circular Motion	$x=cx+r \cdot \cos(\theta+\omega\Delta t), y=cy+r \cdot \sin(\theta+\omega\Delta t)$	ObstacleTracker
Kinematics	2nd-Order Pred.	$P_k = P_0 + V \cdot k\Delta t + \frac{1}{2}A \cdot (k\Delta t)^2$	TrajectoryEstimator
Avoidance	Velocity Blend	$v_{new} = v_{cur} + \alpha(v_{goal} + v_{avoid} - v_{cur})$	AvoidanceController
Avoidance	Accel. Clamp	$\text{if } \Delta v/\Delta t > a_{max}: \Delta v \leftarrow \Delta v \cdot a_{max}/ \Delta v/\Delta t $	AvoidanceController
Avoidance	Collision Cone	$P_{cone} = (1 - \text{angle}/\text{cone_half}) \cdot \text{time_factor}$	DynamicObstaclePredictor
Avoidance	Learned Risk	$\text{risk}_k = 0.86 \cdot \text{risk}_{\{k-1\}} + 0.14 \cdot (\text{speed}/20 + a/6)$	DynamicObstaclePredictor
Acoustic	GCC-PHAT	$\hat{G}(f) = X_i X_j^* / X_i X_j^* ; \hat{\tau} = \text{argmax IFFT}(\hat{G})$	CrossCorrelationEngine
Acoustic	TDOA NLS	$\min \Sigma(d_j - d_i - c \cdot \Delta \tau_{ji})^2, c=343\text{ m/s}$	AcousticFusionEngine
Acoustic	Confidence	$\text{conf} = 1/(1 + \text{RMSE}/6.0)$	AcousticFusionEngine
ML	Poly Features	$\varphi(x) = [1, x_i, x_{ix_j}, x_i^2] \text{ (} d=2 \text{)}$	PhysicalMLTrainer
ML	Pseudoinverse	$W = \Phi^* y = (\Phi^T \Phi)^{-1} \Phi^T y$	PhysicalMLTrainer
ML	Lookahead Target	$P_{new} = P + v_{avoid} \cdot \tau, \tau=1.2\text{ s}$	PathReplanner
Latency	Window Mean	$\mu = (1/N) \Sigma \text{sample}_i \cdot 1000 \text{ [ms]}$	LatencyMonitor
Latency	Jitter (Std Dev)	$\sigma = \sqrt{((1/N) \Sigma (s_i - \mu)^2)} \text{ [ms]}$	LatencyMonitor
Latency	Adaptive Threshold	$\theta_i = \text{clip}(\mu_i \cdot 1.25 + 2.5\sigma_i, 80, 500) \text{ ms}$	SwarmManager
C++ HAL	NED Conversion	$N=R_e(\varphi-\varphi_0)\pi/180, E=R_e(\lambda-\lambda_0)\cos(\varphi_0)\pi/180$	Utils::gpsToNED
C++ HAL	Thrust Redistribution	$T_k = T_{nom} \cdot N_{total}/N_{operational}$	DroneController
Formation	V-Pattern	$\Delta x_i = [i/2] \cdot d \cdot \cos(\pm 45^\circ), \Delta y_i = -[i/2] \cdot d \cdot \sin 45^\circ$	SwarmManager
Formation	Line Pattern	$\Delta x_i = 0, \Delta y_i = -i \cdot d, d=45\text{ m}$	SwarmManager

XVIII. SYSTEM CONSTANTS & CONFIGURATION REFERENCE

Constant	Value	Module	Purpose
<code>Drone.MAX_SPEED</code>	15.0 m/s	drone.py	Maximum drone airspeed
<code>Drone.MAX_ALTITUDE</code>	10,000 m	drone.py	Absolute altitude ceiling
<code>Drone.TAKEOFF_ALTITUDE</code>	120.0 m	drone.py	Default takeoff target altitude
<code>Drone.MAX_OPERATION_RADIUS</code>	10,000 m	drone.py	10 km geofence radius from home
<code>Drone.CRITICAL_BATTERY</code>	20.0 %	drone.py	Emergency landing threshold
<code>Drone.LOW_BATTERY</code>	30.0 %	drone.py	Return-to-home threshold
<code>Drone.BATTERY_IDLE</code>	0.001 %/s	drone.py	Idle discharge rate
<code>Drone.BATTERY_HOVER</code>	0.01 %/s	drone.py	Hover discharge rate
<code>Drone.BATTERY_FLYING</code>	0.02 %/s	drone.py	Flying discharge rate
<code>SwarmManager.HEARTBEAT_TIMEOUT</code>	5.0 s	swarm_manager.py	Drone failure declaration threshold
<code>SwarmManager.ELECTION_TIMEOUT</code>	3.0 s	swarm_manager.py	Leader election window
<code>SwarmManager.dynamic_collision_threshold</code>	0.42	swarm_manager.py	Minimum P_traj to trigger avoidance
<code>SwarmManager.follow_spacing_m</code>	45.0 m	swarm_manager.py	Inter-drone formation spacing
<code>SwarmManager.mission_arrival_threshold_m</code>	6.0 m	swarm_manager.py	Mission completion distance
<code>SwarmManager.acoustic_latency_limit_ms</code>	280.0 ms	swarm_manager.py	Latency gate for full vs. local-only acoustic
<code>SwarmManager.acoustic_confidence_threshold</code>	0.65	swarm_manager.py	Min confidence to dispatch drones
<code>LatencyMonitor.latency_threshold_ms</code>	220.0 ms	latency_monitor.py	Fallback activation threshold
<code>MLBridge.watchdog_timeout_s</code>	1.8 s	latency_monitor.py	Bridge silence duration for watchdog trigger
<code>PBKDF2 iterations</code>	100,000	communication.py	Key stretching iteration count
<code>AES key length</code>	256 bits	communication.py	Symmetric encryption key size
<code>GCM IV length</code>	128 bits	communication.py	Initialization vector size
<code>GCM Tag length</code>	128 bits	communication.py	Authentication tag size
<code>Multicast group</code>	224.0.0.251	communication.py	Local multicast IP group
<code>SPEED_OF_SOUND_MPS</code>	343.0 m/s	acoustic_tracking.py	Speed of sound in air at 20°C
<code>GPS default ref (Dhaka)</code>	23.8103°N, 90.4125°E	drone.py	Default GPS coordinate origin
<code>R_earth</code>	6,371,000 m	drone.py / C++ Utils	Earth radius for flat-Earth model
<code>C++ LatencyMonitor threshold</code>	220.0 ms	dronecontroller.h	C++ fallback trigger threshold
<code>LatencyMonitor window_size</code>	120 samples	latency_monitor.py	Rolling statistics window
<code>Drone auto_motion_radius</code>	220.0 m	drone.py	Auto-hover patrol radius
<code>Drone.LANDING_SPEED</code>	1.0 m/s	drone.py	Descent speed during landing

XIX. REFERENCES

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End of Technical Report — Secure Drone Swarm Management System v1.0.0